Mental Representation and Mental Health Based on Social Media Texts: How Do Temporal Dynamics of Emotional Expression Predict Depression Risk?

Yayun Zheng

"Sometimes I really feel like I can't hold on anymore. Everyone leaves, and no one really understands me."

Problem: Depression often unfolds subtly through emotional shifts in social media posts.

Why social media?

Real-time, natural, emotional expression

More dynamic and open than self-report

What can we do?

Focus on static sentiment, not emotional change

Rare integration of emotion dynamics + mental representation

PSMU and MHL often studied separately

What are we going to do in this study?

Track emotion over time (inertia, amplitude, stability)

Analyze agency/attribution in language

Combine NLP and psychology to detect risk earlier

Theories

Beck's cognitive model, Attribution reformulation theory

- ***Can emotional dynamics predict future depression risk?***
- ***Do mental representations shape how depression manifests linguistically?***
- ***Do PSMU and MHL modulate the effect of emotional patterns on depression?***

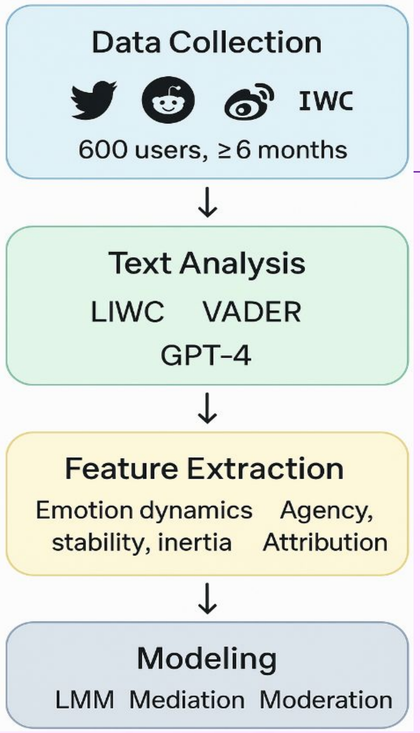
Method

Participants

600 users: 300 with depression, 300 matched controls

Data: X (Twitter), Reddit, Weibo

Activity: ≥6 months



IV:

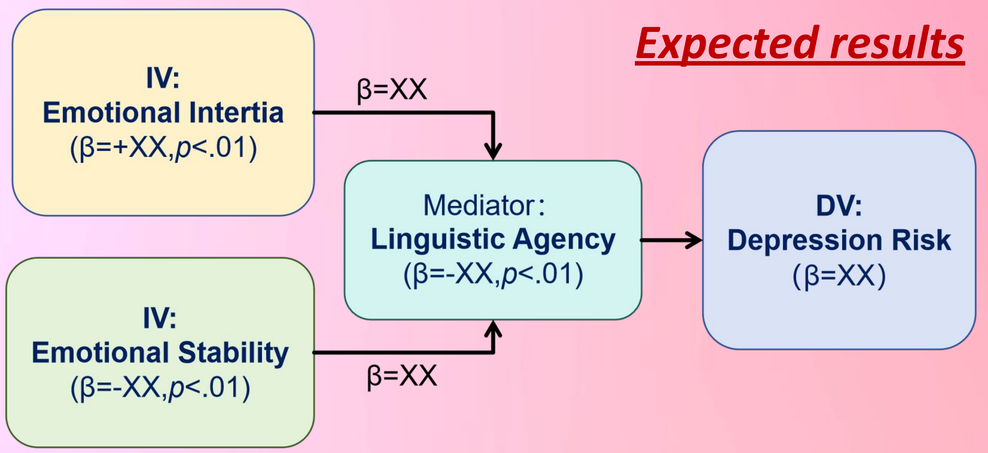
- **Emotion Amplitude:** range of sentiment
- **Emotion Stability**: variability
- **Emotion Inertia:** persistence across time

Mediators / Moderators:

- Agency (e.g., "decide", "choose")
- Attribution (e.g., "always", "my fault")
- Mental Health Literacy (MHL), PSMU

DV: Depression risk (labeled from CLPsych and related corpora)

Expected results



Discussion

- **Theoretical Insight**

Supports cognitive models of depression (e.g., Beck)

Connects affective dynamics to internal representations

- **Applied Value**

Moves from black-box models to interpretable digital biomarkers

Enables early screening using everyday social media expression

-->Can digital systems be designed to promote a stronger sense of agency in online self-expression?